1.FCFS

```
#include <stdio.h>
int main()
{
    int n, i;

    printf("Enter total number of processes: ");
    scanf("%d", &n);

    int at[n], bt[n], ct[n], tat[n], wt[n];

    // Input
    for(i = 0; i < n; i++)
    {
        printf("\nProcess [%d]\n", i+1);

        printf("Arrival time: ");
        scanf("%d", &at[i]);

        printf("Burst time: ");
        scanf("%d", &bt[i]);
    }

    int current_time = 0;

    // FCFS Calculation
    for(i = 0; i < n; i++)
    {
```

```
        if(current_time < at[i])
            current_time = at[i];

        ct[i] = current_time + bt[i];
        tat[i] = ct[i] - at[i];
        wt[i] = tat[i] - bt[i];

        current_time = ct[i];
    }

    float avg_tat = 0, avg_wt = 0;

    // Output Table
    printf("\nProcess Arrival Time Burst Time Completion Time Turnaround Time Waiting
Time\n");

    for(i = 0; i < n; i++)
    {
        printf("%d\t%d\t%d\t%d\t\t%d\t\t%d\n",
        i+1, at[i], bt[i], ct[i], tat[i], wt[i]);

        avg_tat += tat[i];
        avg_wt += wt[i];
    }

    avg_tat /= n;
    avg_wt /= n;

    printf("\nAverage Turnaround Time: %.2f", avg_tat);
```

```
    printf("\nAverage Waiting Time: %.2f\n", avg_wt);


    return 0;
}
```

```
"C:\Users\BMSCECSE-l4\Desk

Enter total number of processes: 3

Process [1]
Arrival time: 0
Burst time: 6

Process [2]
Arrival time: 3
Burst time: 8

Process [3]
Arrival time: 7
Burst time: 2

Process Arrival Time Burst Time Completion Time Turnaround Time Waiting Time
1       0       6       6               6               0
2       3       8       14              11              3
3       7       2       16              9               7

Average Turnaround Time: 8.67
Average Waiting Time: 3.33

Process returned 0 (0x0)   execution time : 28.683 s
Press any key to continue.
```

2.SJF Non Pre-emptive

```
#include <stdio.h>

int main()
{
    int n, i, j;
    printf("USN: 1WA24CS198");
    printf("\nEnter number of processes: ");
    scanf("%d", &n);

    int at[n], bt[n], ct[n], tat[n], wt[n];
    int finished[n];

    for(i = 0; i < n; i++)
        finished[i] = 0;

    printf("Enter arrival times:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &at[i]);

    printf("Enter burst times:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &bt[i]);

    int current_time = 0, completed = 0;
    int min_index;
```

```
while(completed < n)
{
    min_index = -1;
    int min_bt = 10000;

    for(i = 0; i < n; i++)
    {
        if(at[i] <= current_time && finished[i] == 0)
        {
            if(bt[i] < min_bt)
            {
                min_bt = bt[i];
                min_index = i;
            }
        }
    }

    if(min_index == -1)
    {
        current_time++;
    }
    else
    {
        ct[min_index] = current_time + bt[min_index];
        tat[min_index] = ct[min_index] - at[min_index];
        wt[min_index] = tat[min_index] - bt[min_index];

        current_time = ct[min_index];
```

```
            finished[min_index] = 1;
            completed++;
        }
    }

    float avg_tat = 0, avg_wt = 0;

    printf("\nSJF scheduling:\n");
    printf("PID\tAT\tBT\tCT\tTAT\tWT\n");

    for(i = 0; i < n; i++)
    {
        printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
        i+1, at[i], bt[i], ct[i], tat[i], wt[i]);

        avg_tat += tat[i];
        avg_wt += wt[i];
    }

    avg_tat /= n;
    avg_wt /= n;

    printf("\nAverage turnaround time:%fms\n", avg_tat);
    printf("\nAverage waiting time:%fms\n", avg_wt);

    return 0;
}
```

```
C:\Users\BMSCECSE-I4\Deskt

USN: 1WA24CS184
Enter number of processes: 5
Enter arrival times:
0
3
6
8
4
Enter burst times:
6
4
9
2
1

SJF scheduling:
PID     AT      BT      CT      TAT     WT
P1      0       6       6       6       0
P2      3       4       11      8       4
P3      6       9       22      16      7
P4      8       2       13      5       3
P5      4       1       7       3       2

Average turnaround time:7.600000ms

Average waiting time:3.200000ms

Process returned 0 (0x0)   execution time : 22.610 s
Press any key to continue.
```

3.SRTF

```
#include <stdio.h>

int main()
{
    int n, i, time = 0, completed = 0, min_index;
    printf("USN: 1WA24CS198");
    printf("\nEnter the number of processes: ");
    scanf("%d", &n);

    int at[n], bt[n], rt[n], ct[n], tat[n], wt[n];

    for(i = 0; i < n; i++)
    {
        printf("Enter arrival time and burst time for process %d: ", i+1);
        scanf("%d %d", &at[i], &bt[i]);
        rt[i] = bt[i]; // remaining time
    }

    while(completed < n)
    {
        min_index = -1;
        int min_rt = 10000;

        for(i = 0; i < n; i++)
        {
            if(at[i] <= time && rt[i] > 0)
```

```
        {
            if(rt[i] < min_rt)
            {
                min_rt = rt[i];
                min_index = i;
            }
        }
    }

    if(min_index == -1)
    {
        time++;
    }
    else
    {
        rt[min_index]--;
        time++;

        if(rt[min_index] == 0)
        {
            completed++;
            ct[min_index] = time;
            tat[min_index] = ct[min_index] - at[min_index];
            wt[min_index] = tat[min_index] - bt[min_index];
        }
    }
}
```

```
    float avg_wt = 0, avg_tat = 0;

    printf("\nPID\tAT\tBT\tCT\tTAT\tWT\n");

    for(i = 0; i < n; i++)
    {
        printf("%d\t%d\t%d\t%d\t%d\t%d\n",
        i+1, at[i], bt[i], ct[i], tat[i], wt[i]);

        avg_wt += wt[i];
        avg_tat += tat[i];
    }

    avg_wt /= n;
    avg_tat /= n;

    printf("\nAverage Waiting Time: %.2f\n", avg_wt);
    printf("Average Turnaround Time: %.2f\n", avg_tat);

    return 0;
}
```

```
C:\Users\BMSCECSE-l4\Deskt

USN: 1WA24CS184
Enter the number of processes: 5
Enter arrival time and burst time for process 1: 5
9
Enter arrival time and burst time for process 2: 8
6
Enter arrival time and burst time for process 3: 3
1
Enter arrival time and burst time for process 4: 0
5
Enter arrival time and burst time for process 5: 2
4

PID     AT      BT      CT      TAT     WT
1       5       9       25      20      11
2       8       6       16      8       2
3       3       1       4       1       0
4       0       5       6       6       1
5       2       4       10      8       4

Average Waiting Time: 3.60
Average Turnaround Time: 8.60

Process returned 0 (0x0)   execution time : 32.728 s
Press any key to continue.
```